

- Snapdragons, 279f
 Snapping shrimp, 512f, 513
 Snook, Rhonda, 1024f
 Snow geese, 1256f–1257f
 Snow pea, 650f
 Snowshoe hares (*Lepus americanus*), 501f, 1205f
 Soapberry bugs, 477f, 478, 1187
 Social behavior, 1159f, 1160
 Social learning, **1147f**–1148f, 1153
 Social play, 1159
 Society
 plant biotechnology and, 840
 population age structure and, 1208
 Society, science and, 19f, 23, 24f
 Sociobiology, 1159–1160
Sociobiology: The New Synthesis (book), 1159
 Sockeye salmon (*Oncorhynchus nerka*), 978f
 Sodium, 29t
 Sodium chloride
 elimination of excess, by marine birds, 982f
 human diets and, 901
 kidney processing of, 988f–990f, 991
 osmoregulation of, 992, 993f
 plant responses to excessive, 865
 as table salt, 29f, 38f, 49f
 in treating diarrhea, 139
 Sodium ions, 1070t, 1071f
 Sodium-potassium pump, **137**, **1069**–1070
 as active transport, 136, 137f
 neuron resting potential and, 1070f–1071f
 ouabain and, 1084
 Software, bioinformatics, 444, 445f–448f
 Soil
 acid precipitation and, 821
 antibiotics from bacteria in, 590
 bacteria in, 584f, 586f, 814f–816f, 817
 bryophyte reduction of nitrogen leaching from, 627f
 determining diversity of bacteria in, 1223f
 fossils in, 529f
 fungi in, 656f
 limiting nutrients in, 1244
 plant resource acquisition from, 787
 plant response to excessive salt in, 142, 865
 species distributions and, 1186
 sustainable agriculture and conservation of, 807f–809f
 texture and composition of, 806f–807f
 Soil conservation, 807f–809f
 Soil horizons, **806f**
 Soil worm model organism. *See* *Caenorhabditis elegans*
 Solar cycle, 1282
 Solar energy
 absorption of, determining primary production with, 1242f
 in energy flow and chemical cycling, 9f, 1240f
 global energy budget and, 1241
 in photosynthesis, 41f, 187f
 primary production in aquatic ecosystems and limitations of, 1242
Solidago canadensis, 812
 Solute potential, **789**, 804
 Solute potential equation, 804
 Solutes, **49**
 animal osmoregulation of, 981, 982f
 diffusion of, 132, 133f
 diffusion of, across capillary walls, 932, 933f
 effects of, on water potential, 789–790
 ion gradients and transport of, 993f
 kidney gradients of, 989, 990f, 991
 osmoregulation of, 977f–982f
 short-distance transport of, across plant plasma membranes, 788, 789f
 solvents and, 49f, 50
 transport of, from roots to shoots via xylem, 792, 793f–795f, 796
 transport of, from sources to sinks via phloem, 799, 800f–801f
 Solutions, **49f**, 50
 Solvents, **49f**, 50
 Somatic cells, **236**, **255**
 Somatosensory cortex, 1097f, 1098
 Somites, 720, 1054f, **1055**
 Songs, bird learning of, 1147
 Sonic hedgehog growth factor, 1063–1064
 Soredia, **668**, 669f
 Sorghum, 813f
 Sori, **631**
 Soufriere Hills volcano, 551
 Sound receptors, 1110, 1112f–1116f
 Sour tastes, 1123f–1124f
 South Africa, restoration projects in, 1254f
 South America, 743–744
 South American vampire bats, 991f, 992
 South Atlantic Subtropical Gyre, 1168f
 Southern flying squirrel (*Glaucomys volans*), 517
 Southern stingray (*Dasyatis americana*), 727f
 South Pacific Subtropical Gyre, 1168f
 Soybeans, 211, 493, 815f–816f, 817, 1015
 Space-filling models, 36, 37f, 39f, 59f, 74f, 79f, 319f
 Spain, 573f
 Spanish flu pandemic, 410
Spartina patens, 1186
 Spatial learning, 1144, **1145f**
 Spatial summation, 1079f
 Spawning, 1022–1023
 Speciation, **507**
 allopatric, 511f–512f, 513
 allopatric vs. sympatric, 511f, 516
 as conceptual bridge between microevolution and macroevolution, 507, 523
 Darwinian theory of, 473, 474f–476f
 genetics of, 522f, 523
 of humans, 524
 hybrid zones and reproductive isolation in, 516, 517f–519f, 520
 as origin of species in Darwinian evolution, 506f, 507
 orthologous genes and, 566
 reproductive isolation and biological species concept in, 507f–510f
 species selection and differential, 548–549
 sympatric, 513, 514f–515f, 516
 time course of, 520f–521f, 522
 Speciation clock, 522
 Species, **507**
 biological concept of, 507f–510f
 classification of, 12f–13f, 470, 554, 555f
 communities of. *See* Communities
 comparing developmental processes of, 463f–464f
 comparing genomes of, 459, 460f–462f
 complete genome sequences for, 448t
 cross-species gene expression in evolution of, 423
 C. Darwin's theory of origin and evolution of, 14f–16f, 473, 474f–476f. *See also* Evolution; Speciation
 discovery of new, 1164f, 1185, 1260f
 distributions of. *See* Species distributions
 diversity of. *See* Species diversity
 edge, 1271
 endangered or threatened, 1261, 1262f, 1272f, 1288
 extinction of mollusc, 702f
 foundation, 1214, 1226
 fusion of, 519f, 520
 genome size, number of genes, gene density, and noncoding DNA for, 448t, 449
 geographic distribution of, 482f, 483
 homologous genes in, 565f, 566
 interactions between. *See* Interspecific interactions
 interactions of, 1214f–1221f, 1237
 introduced, 1223, 1264f, 1287
 keystone, 1226f, 1270
 large-impact, 1225, 1226f
 loss of amphibian, 733
 metagenomics and genome sequencing of groups of, 444
 morphological and ecological concepts of, 507, 508f–509f, 510
 number of named, 1261
 phylogenies as history of, for species, 554f–556f. *See also* Phylogenies
 populations of. *See* Populations
 resurrection of extinct, 1266
 in tree of life, 15f–16f
 tropical deforestation and extinctions of, 651f, 652
 using phylogenetic gene trees to identify, of whale meat, 557f
 Species-area curve, **1232f**
 Species distributions
 abiotic factors in, 1178, 1183f–1186f
 in aquatic biomes, 1178
 biotic factors in, 1178, 1183f–1184f
 climate and, 1164, 1170f
 determinants of, 1164f
 dispersal in, 1183, 1184f
 evolution and, 1183, 1189
 natural range expansion, 1183, 1184f
 Species diversity, **1222f**
 benefits of, to humans, 1262, 1263f
 biodiversity crisis and, 1260f–1262f
 biogeographical factors affecting, 1231, 1232f–1233f
 bottom-up and top-down controls in, 1226, 1227f
 climate change effects on, 1279
 community stability and, 1223f
 disturbances influencing, 1228, 1229f–1231f
 human impacts on, 1231f
 large-impact species in, 1225, 1226f
 as level of biodiversity, 1261f–1262f
 protection of, 1260
 resurrection of, 1266
 species richness and relative abundance in, 1222f
 sustainable development and, 1284, 1285f, 1286
 threats to, 1260f, 1261, 1263, 1264f–1266f
 trophic structure and, 1223, 1224f–1227f. *See also* Trophic structure
 Species richness, **1222**
 biogeographic factors affecting, 1231, 1232f–1233f
 species-area curves of, 1232f
 in species diversity, 1222f. *See also* Species diversity
 Species selection, 548–549
 Species transplants, 1184
 Specific heat, **46**, 47f
 Specificity
 cell-signaling, 227, 228f
 enzyme substrate, 155f, 156
 polymerase chain reaction, 421, 422f
 viral, 401
 Specific transcription factors, 374f–375f, 845
 Spectrophotometer, **193f**
 Speech
 brain function and, 1098f
 FOXP2 gene and, 461, 462f
 Spemann, Hans, 1060, 1061f–1062f
 Sperm, **1020**
 biased usage of, in female fruit flies, 1024f
 chromosomes in human, 236–237
 conception and, 1035f
 in fertilization, 1022f–1024f, 1044f, 1045
 flagellated, in plants, 619
 human spermatogenesis and, 1027, 1028f, 1031f, 1042
 mammalian sex determination and, 298f
 seed plant. *See* Pollen grains
 Spermathecae, 711, **1024f**
 Spermatids, 1028f
 Spermatocytes, 1028f
 Spermatogenesis, **1027**, 1028f, 1031f, 1042
 Spermatogonia, **1028f**
 Spermicidal foam or jelly, 1038f
Spermophilus parryii, 893
Sphagnum moss, **627f**, 628
 S phase, **237f**
Sphenodon punctatus, 737f
 Spherical prokaryotes, 574f
 Sphincters, 906f, **907**
 precapillary, 932f
 Sphygmomanometer, 931f
 Spiders, 45f, 707, 708f, 951
 Spikemosses, 631, 632f, 633
Spilogale species, 508f
 Spina bifida, 1055
 Spinal cord, 1086f–1089f
 Spines, 762f
 Spines, sea star, 714f
 Spinnerets, 708
 Spiny-headed worms, 687f
 Spiny mouse, 1216f
 Spiral cleavage, **681**, 682f
 Spiral phyllotaxy, 786f
 Spiral prokaryotes, 574f
 Spiral valve, shark, 727
Spirobranchus giganteus, 703f
 Spirochetes, 584f
Spirodela oligorrhiza, 101f